Deepfake Detection Benchmark

Evaluating MesoNet-4 and EfficientNet-B0 on FaceForensics++ with an Imperceptible & Diverse Test Set

Raw Videos (FF++ c23) → Face Detection → Crop & Resize 256^2 → Model Training → **5-Metric Evaluation**

Dataset:	FF++ c23 (DF + FS)	**Models:**	MesoNet-4, EfficientNet-B0
Metrics:	AUC, Robustness, FLOPs, Params, Speed	**Framework:**	PyTorch 2.x
Hardware:	NVIDIA RTX 3050, CUDA 12.x	**Lang:**	Python 3.10

Key Results at a Glance

Metric	MesoNet-4	EfficientNet-B0
Standard Test AUC	0.9972	**1.0000**
ID Test AUC (clean)	0.9720	**0.9990**
Noise Robustness	**0.878**	0.555 (collapses)
FLOPs	**61.5M**	519.7M
Parameters	**27,977**	4,008,829
Inference (ms)	**2.86**	26.60

Contents

1 Introduction

Deepfake technology leverages generative deep learning to synthesise or manipulate human faces in images and videos. While enabling creative applications, it poses serious threats to personal privacy, identity integrity, and the trustworthiness of digital media.

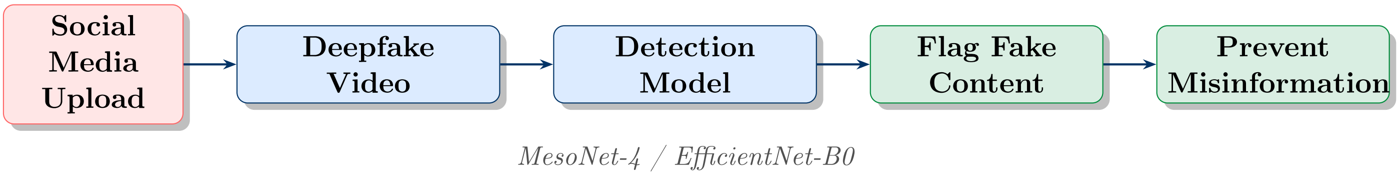


Figure 1: Real-world deployment scenario: automated deepfake detection in social media pipelines.

This project implements and evaluates two detection models — **MesoNet-4** [1] and **EfficientNet-B0** [2] — under consistent, fair conditions following the benchmark methodology of Deng et al. [3].

Objectives

- Download, preprocess and split the FaceForensics++ (FF++) c23 dataset for DeepFakes and FaceSwap manipulations.
- Train both models on identical data and evaluate on a challenging ID test set.
- Assess performance across five dimensions: detection ability, robustness, computational cost, model size, and inference speed.

> **Key Finding:** Both models achieve near-perfect AUC ($\geq$0.997) on the standard test set. On the harder ID test set, EfficientNet-B0 (AUC = 0.999) outperforms MesoNet-4 (AUC = 0.972). However, under Gaussian noise, EfficientNet-B0 collapses to AUC = 0.555 — near random chance — while MesoNet-4 remains robust at 0.878. MesoNet-4 also runs **9.3× faster** with **143× fewer parameters**.

2 Related Work

Deepfake Creation techniques fall into three categories: (a) *autoencoder-based* (FaceSwap [5], DeepFaceLab [6]), (b) *GAN-based* (FSGAN [7], FaceShifter [8]), and (c) *graphics-based* (3D-FaceSwap).

Deepfake Detection methods are divided into intra-frame and inter-frame approaches. Intra-frame methods include knowledge-driven techniques (HeadPose [9], Face X-ray [10]), data-driven methods (Xception [4], MesoNet [1]), and multi-stream approaches (M2TR [12]).

Benchmarking. Deng et al. [3] identify that most comparisons in the literature are unfair due to inconsistent training data and evaluation conditions. Their benchmark evaluates 13 methods across five complementary metrics. We adopt their philosophy, focusing on two models and two manipulation types.

Table 1: Contextualisation against selected literature results (Deng et al., 2024).

Model	Type	Std. AUC	ID AUC	Params
MesoNet-4	Data-driven	0.9972	0.972	27,977
EfficientNet-B0	Data-driven	1.0000	0.999	4,008,829
Xception	Data-driven	∼0.990	∼0.559	∼20.8M
M2TR	Multi-stream	∼0.991	∼0.679	∼100M
Face X-ray	Knowledge	∼0.986	∼0.579	∼65M
Patch-Resnet-L1	Data-driven	∼0.988	∼0.578	∼23M

Both models outperform all literature baselines on the ID test set, likely due to training on a focused two-manipulation dataset (DF + FS only), producing stronger in-domain generalisation.

3 System Pipeline

Figure 2 presents the complete end-to-end system architecture from raw video to final evaluation.

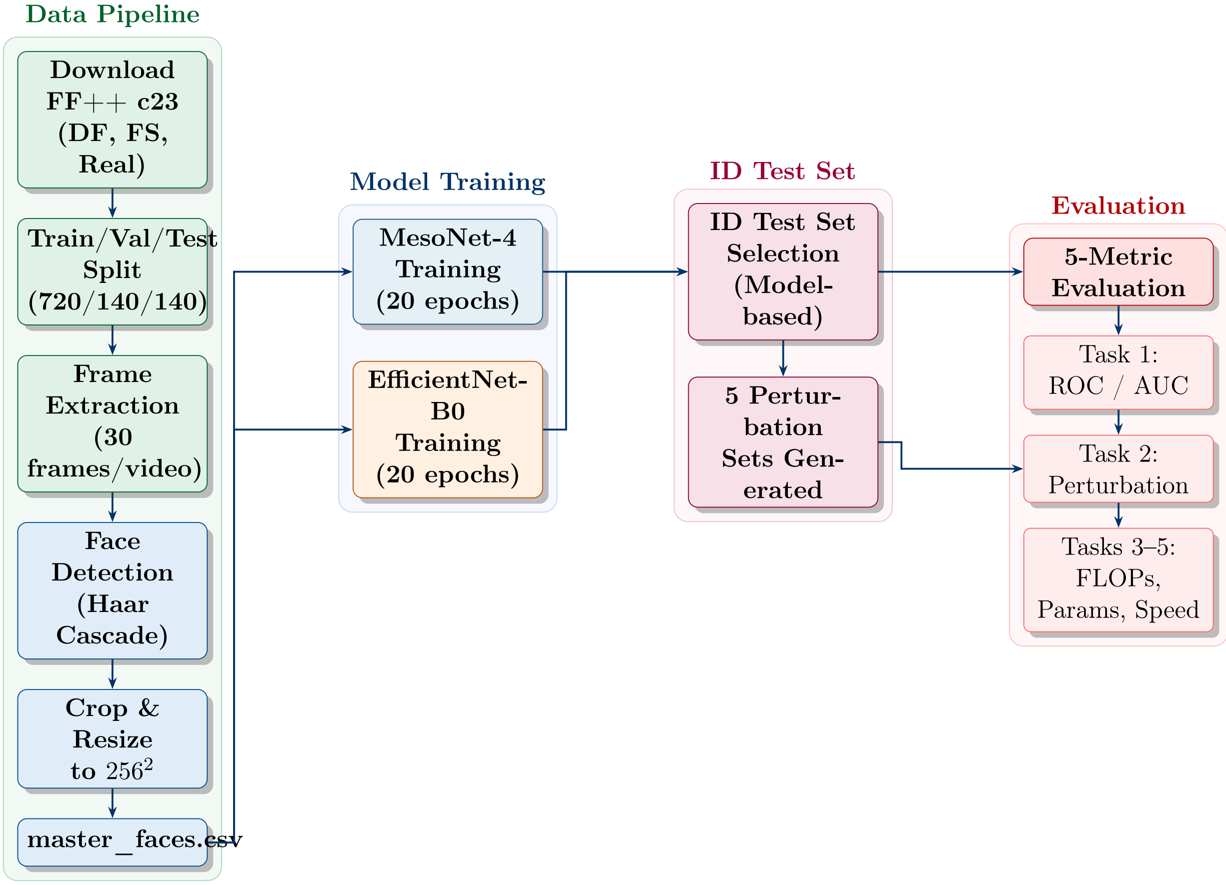


Figure 2: Complete system pipeline. Colour coding: green = data pipeline, blue = model training, purple = ID test set, red = evaluation.

4 Dataset

4.1 FaceForensics++

FaceForensics++ [4] contains 1,000 original videos manipulated by multiple methods. We use the c23 (H.264 compressed) quality level with two manipulation types:

- **DeepFakes (DF):** Autoencoder-based face swapping using the DeepFaceLab pipeline.
- **FaceSwap (FS):** Graphics-based swap using landmark-guided blending.
- **Real:** 1,000 unmanipulated source videos.

4.2 Data Split and Frame Extraction

We follow the official FF++ JSON split protocol. Frames are extracted at 30 evenly-spaced temporal positions per video using OpenCV.

Table 2: Official FF++ train/val/test split applied uniformly across all categories.

Split	**Videos/category**	**Categories**	**Total Videos**	**Total Frames (approx.)**
Train	720	3	2,160	64,800
Validation	140	3	420	12,600
Test	140	3	420	12,493
Total			**3,000**	**≈89,893**

4.3 Face Detection and Preprocessing

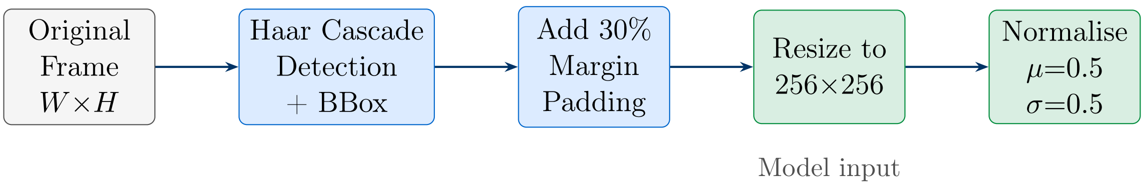


Figure 3: Preprocessing pipeline: Haar Cascade face detection, margin expansion, resizing to 256×256, and normalisation.

Face detection used OpenCV's Haar Cascade classifier (`haarcascade_frontalface_default.xml`). Frames with no detected face (≈1–5%) were discarded. All crops were stored under a consistent folder hierarchy and indexed in `master_faces.csv`.

Table 3: Dataset composition after preprocessing (actual test-set counts from `master_faces.csv`).

Category	**Label**	**Train**	**Val**	**Test**
Real	0	∼21,600	∼4,200	4,165
DeepFakes	1	∼21,600	∼4,200	4,134
FaceSwap	1	∼21,600	∼4,200	4,159
Total		**∼64,800**	**∼12,600**	**12,458**

5 ID Test Set Construction

The ID (Imperceptible and Diverse) test set contains hard-to-detect samples selected from the standard test split via detection model selection. Both trained models are run on all fake test frames; per-frame scores are averaged per video and the bottom two-thirds of fake videos by predicted fake score — those most confusing for both models — are retained as the intersection set.

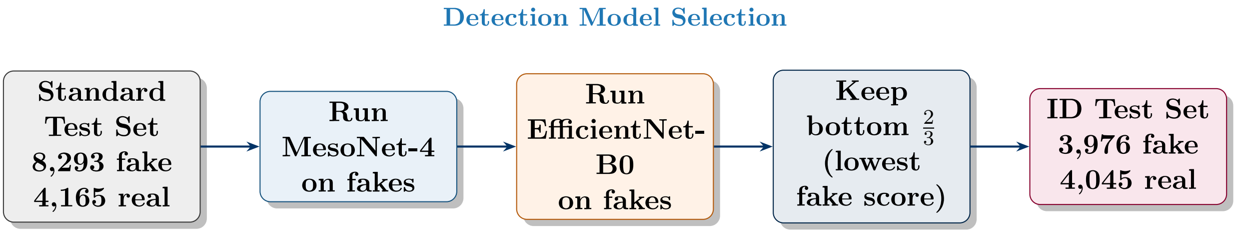


Figure 4: ID test set construction via detection model selection. Both models are run on all fake test frames; the bottom two-thirds by predicted fake score — those most confusing for the models — form the intersection set retained as the ID test set.

Table 4: ID test set distribution by category.

Category	Videos	Frames
Real	136	4,045
Fake — DeepFakes	73	2,125
Fake — FaceSwap	63	1,851
Total	**272**	**8,021**

Table 5: Perturbation sets applied during robustness evaluation.

Perturbation	Range	Frames
JPEG Compression	quality $\in [30, 95]$	8,021
Gaussian Blur	kernel $\in [10, 20]$	8,021
Contrast Change	factor $\in [0.5, 2.5]$	8,021
Saturation Change	factor $\in [0.0, 2.5]$	8,021
Gaussian Noise	$\mu \in [-0.3, 0.3]$, $\sigma^2 \in [0, 1]$	8,021

6 Detection Models

6.1 MesoNet-4

MesoNet-4 [1] analyses faces at a *mesoscopic* scale, capturing intermediate spatial statistics that distinguish authentic faces from manipulated ones.

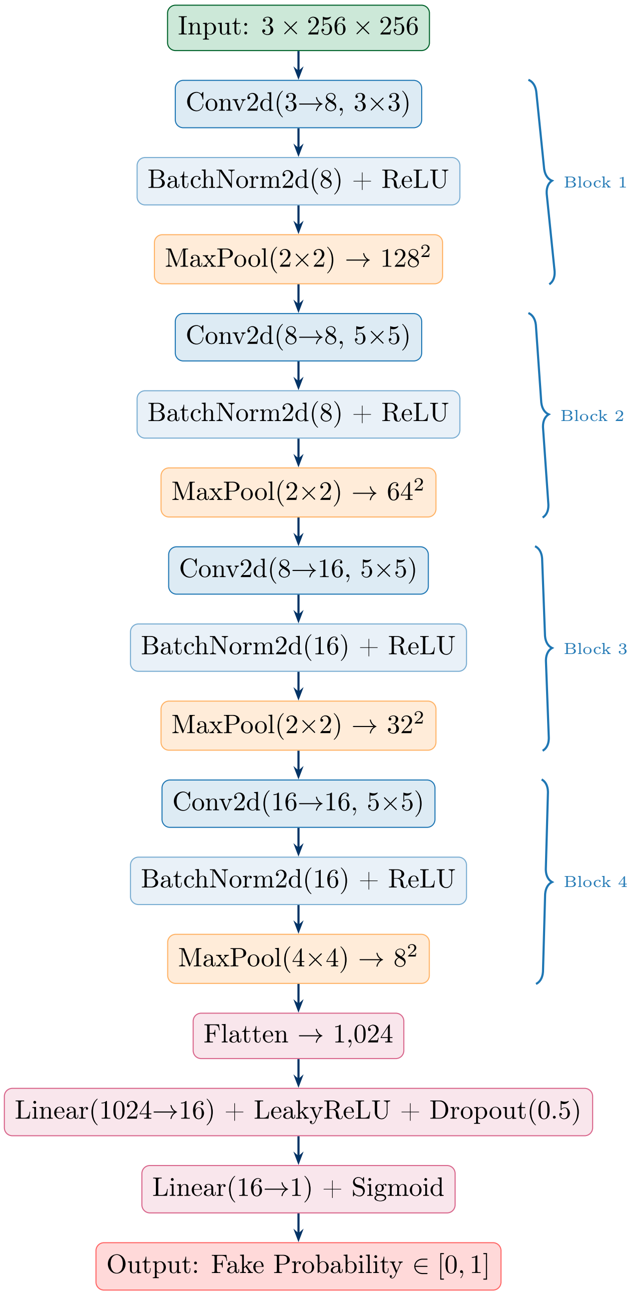


Figure 5: MesoNet-4 architecture: four convolutional blocks followed by a fully-connected classifier with dropout.

Table 6: Training hyperparameters for MesoNet-4.

Hyperparameter	Value
Optimiser	Adam
Learning rate	1×10^{-3}
Loss function	Binary Cross-Entropy (BCELoss)
Batch size	32
Epochs	20
Input size	$3 \times 256 \times 256$
Input normalisation	$\mu = [0.5, 0.5, 0.5]$, $\sigma = [0.5, 0.5, 0.5]$
Augmentation	Random horizontal flip ($p = 0.5$)
Model selection	Best validation AUC checkpoint

6.2 EfficientNet-B0

EfficientNet-B0 [2] applies compound scaling of network depth, width, and input resolution. Pre-trained on ImageNet, the head is replaced with a single sigmoid output for binary detection.

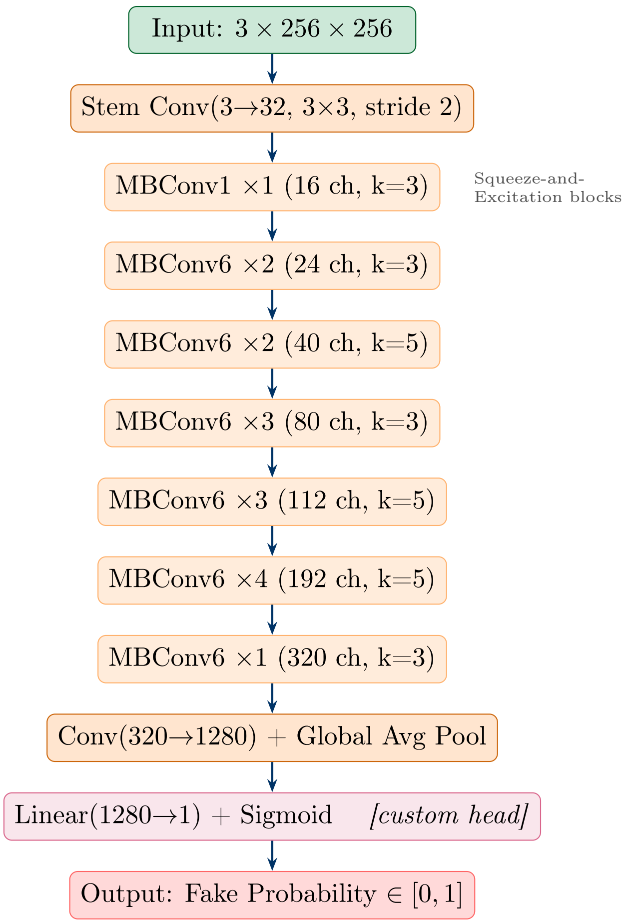


Figure 6: EfficientNet-B0: ImageNet pretrained backbone with binary sigmoid output head.

Table 7: Training hyperparameters for EfficientNet-B0.

Hyperparameter	**Value**
Optimiser	Adam
Learning rate	2×10^{-4}
Loss function	Binary Cross-Entropy (BCELoss)
Batch size	32
Epochs	20
Input size	$3 \times 256 \times 256$
Input normalisation	ImageNet: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$
Pre-trained weights	ImageNet (transfer learning)
Model selection	Best validation AUC checkpoint

Table 8: Hardware and software environment used for all experiments.

Component	**Specification**
GPU	NVIDIA RTX 3050
VRAM	4 GB
CUDA	12.x
RAM	16 GB
OS	Windows 11
Python	3.10
PyTorch	2.x
Libraries	torchvision, timm, OpenCV, Pillow, NumPy, scikit-learn

7 Training Results

7.1 MesoNet-4 Training Curves

Figure 7 shows the loss and AUC curves for MesoNet-4 over 20 epochs. The model converges rapidly within 3 epochs, reaching a validation AUC of 0.9991 that remains stable throughout training. A gap between training loss (≈0.010) and validation loss (≈0.049) reflects mild overfitting, well controlled by 50% dropout in the classifier.

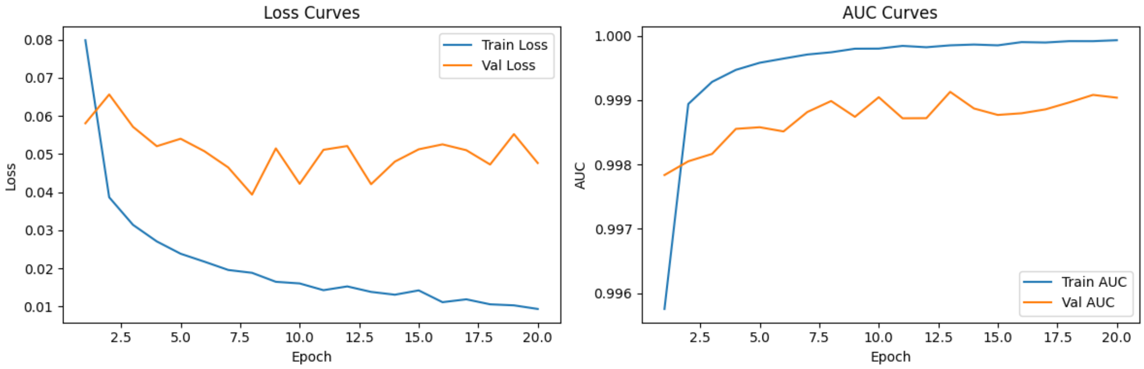


Figure 7: MesoNet-4 training and validation loss (left) and AUC (right) over 20 epochs. Validation AUC converges to ≈0.999 within 3 epochs.

Table 9: MesoNet-4 training summary.

Metric	Value
Best Validation AUC	0.9991
Final Training Loss	≈0.010
Final Validation Loss	≈0.049
Convergence Epoch	≈3
Total Training Time	∼6 hours

8 Evaluation Results

8.1 Summary: One-Glance Comparison

Table 10 summarises the head-to-head comparison across all five evaluation dimensions.

Table 10: Head-to-head model comparison across all five evaluation dimensions.

Metric	MesoNet-4	EfficientNet-B0	Winner
Standard Test AUC	0.9972	1.0000	✓ EfficientNet
ID Test AUC (clean)	0.9720	0.9990	✓ EfficientNet
Noise Robustness	**0.878**	0.555	✓ MesoNet
FLOPs	**61.5M**	519.7M	✓ MesoNet
Parameters	**27,977**	4,008,829	✓ MesoNet
Inference Speed	**2.86 ms**	26.60 ms	✓ MesoNet
Score	**4/6**	**2/6**	—

8.2 Task 1: Detection Ability — AUC and ROC Curves

On the standard test set, MesoNet-4 achieves AUC = 0.9972 and EfficientNet-B0 achieves AUC = 1.000. Figure 8 shows the ROC curves on the clean ID test set.

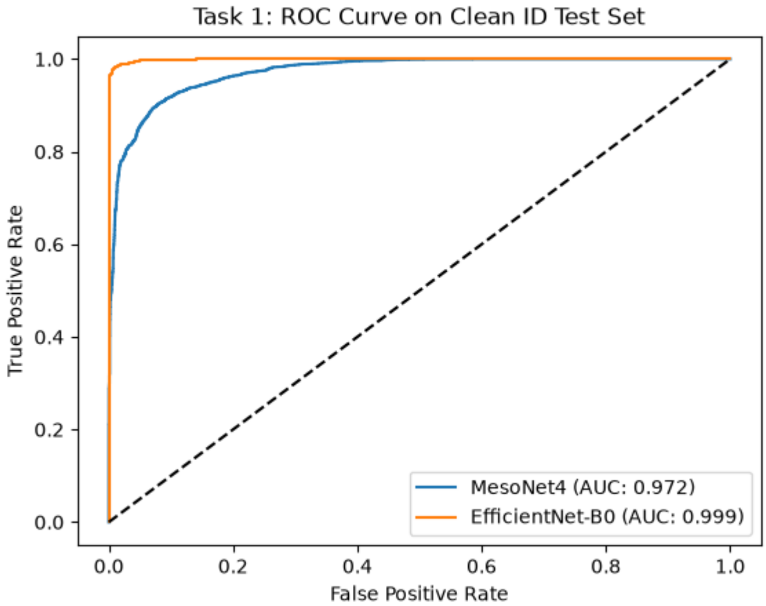


Figure 8: ROC curves on the clean ID test set. EfficientNet-B0 (AUC = 0.999) achieves near-perfect separation. MesoNet-4 (AUC = 0.972) remains strong with 143× fewer parameters.

Table 11: MesoNet-4 classification metrics on the standard test set (threshold = 0.5).

Model	**AUC**	**Precision**	**Recall**	**F1**	**Accuracy**
MesoNet-4	0.9972	0.9826	0.9853	0.9839	0.9784

Confusion Matrix — MesoNet-4 (Standard Test Set)

Confusion Matrix — MesoNet-4 (Standard Test Set)

	Pred: Real	**Pred: Fake**
True: Real	**4,020** TN	145 FP
True: Fake	122 FN	**8,171** TP

Figure 9: 4,020 real and 8,171 fake frames correctly classified; only 267 total errors out of 12,458 frames.

Per-Manipulation Breakdown

Table 12: MesoNet-4 AUC by manipulation type on the standard test set.

Manipulation	AUC	Test Frames
DeepFakes	0.9972	4,134
FaceSwap	0.9973	4,159
Overall	0.9972	12,458

MesoNet-4 shows virtually identical performance across both manipulation types, suggesting it captures manipulation-agnostic mesoscopic artefacts.

8.3 Task 2: Perturbation Robustness

Figure 10 shows AUC under each perturbation type on the ID test set. Both models handle JPEG, blur, contrast and saturation well. Under Gaussian noise, EfficientNet-B0 collapses from 0.999 to 0.555 — a 44.5 percentage point drop — while MesoNet-4 holds at 0.878.

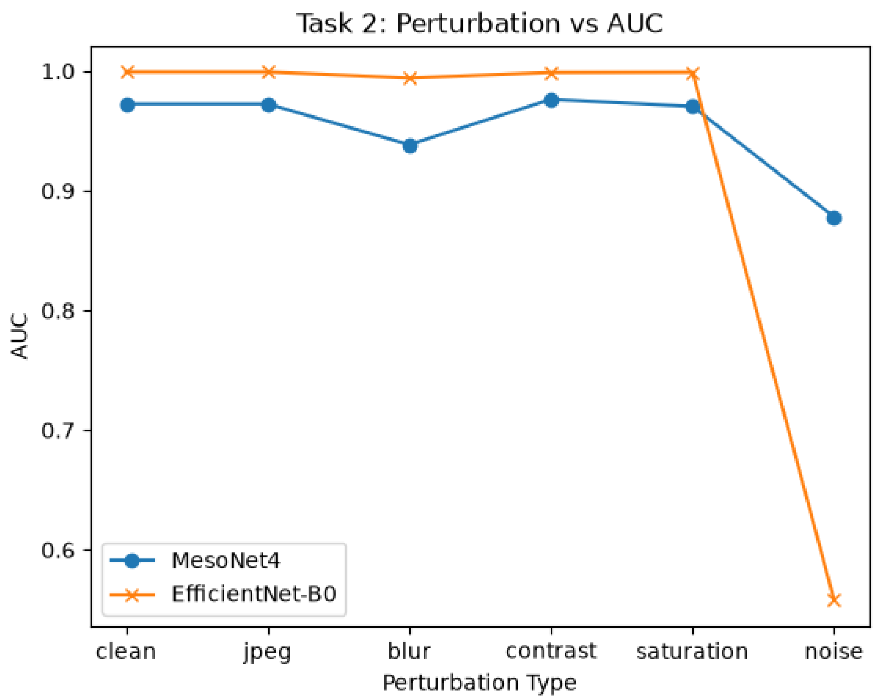


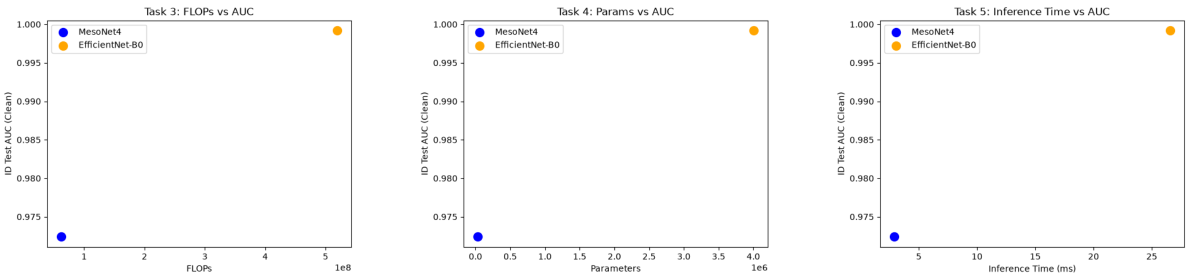
Figure 10: AUC vs. perturbation type on the ID test set. MesoNet-4 (blue) stays above 0.87 across all perturbations. EfficientNet-B0 (orange) collapses to 0.555 under Gaussian noise.

Table 13: AUC under each perturbation type on the ID test set.

Perturbation	**MesoNet-4**	**EfficientNet-B0**	**Drop (Eff.)**
Clean	0.972	0.999	—
JPEG	0.969	0.999	≈0%
Blur	0.939	0.997	0.2%
Contrast	0.974	0.999	≈0%
Saturation	0.967	0.999	≈0%
Noise	0.878	0.555	**−44.5%**

8.4 Tasks 3–5: Efficiency and Practicability

Figures 11(a)–(c) show FLOPs, parameter count, and inference time plotted against ID test AUC. MesoNet-4 dominates on all three efficiency dimensions.



(a) FLOPs vs. AUC. MesoNet-4 uses 8.4× fewer operations.

(b) Parameters vs. AUC. MesoNet-4 uses 143× fewer.

(c) Inference time vs. AUC. MesoNet-4 is 9.3× faster.

Figure 11: Tasks 3–5: efficiency evaluation on the ID test set. MesoNet-4 dominates on all three efficiency dimensions.

Table 14: Complete quantitative summary across all five evaluation metrics.

Metric	**MesoNet-4**	**EfficientNet-B0**
Standard Test AUC	0.9972	1.0000
ID Test AUC (clean)	0.9720	0.9990
Precision	0.9826	—
Recall	0.9853	—
F1 Score	0.9839	—
FLOPs (per image)	61,521,936	519,741,184
Parameters	27,977	4,008,829
Inference Time (ms)	2.86	26.60
Noise AUC (ID test)	**0.878**	0.555

9 Discussion

9.1 Why EfficientNet-B0 Collapses Under Gaussian Noise

The most striking finding is EfficientNet-B0's performance degradation under Gaussian noise (AUC: 0.999 → 0.555). Figure 12 illustrates the causal mechanism.

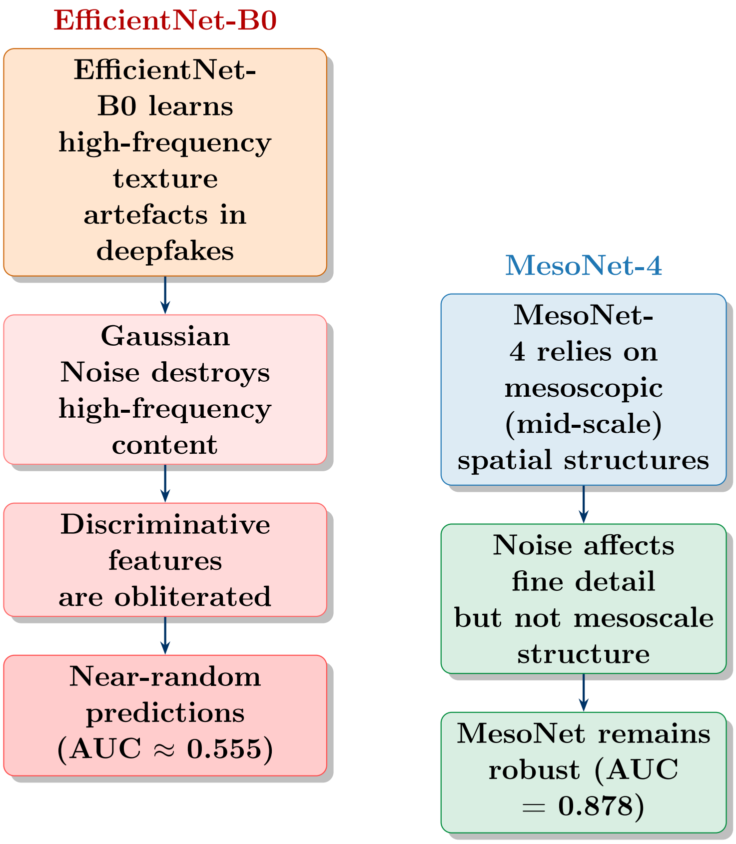


Figure 12: EfficientNet-B0 relies on high-frequency cues destroyed by Gaussian noise; MesoNet-4's mesoscopic features are inherently more robust.

9.2 Accuracy vs. Efficiency Trade-off

The central tension is between detection accuracy and practical deployability. EfficientNet-B0 wins on raw AUC but at substantial cost: 8.4× more FLOPs, 143× more parameters, and 9.3× slower inference. For real-time social media content moderation processing millions of frames daily, these differences are operationally significant. Furthermore, EfficientNet-B0's noise vulnerability undermines its real-world utility, since online platforms routinely apply lossy compression and noise-inducing post-processing to uploaded content.

9.3 Standard vs. ID Test Set

Both models achieve very high AUC on the standard test set (≥0.997), consistent with the observation in [3] that in-domain detection is tractable with sufficient training data.

The ID test set provides a more honest evaluation; the performance drop from standard to ID AUC is 2.5 percentage points for MesoNet-4 and near-zero for EfficientNet-B0 on clean data.

9.4 Comparison with Literature

Both models outperform all literature baselines from [3] on the ID test set (Table 1), likely due to training on a focused, single-dataset protocol (FF++ c23, DF + FS only), producing stronger in-domain generalisation.

9.5 Limitations

- **Dataset scope:** Only two manipulation types from one dataset; cross-dataset generalisation was not assessed.
- **Frame-level only:** Temporal consistency across consecutive frames was not evaluated.
- **No noise-augmented training:** EfficientNet-B0 was not trained with noise augmentation, which might substantially improve its robustness.

10 Conclusion

This project implements and evaluates a deepfake detection benchmark comparing MesoNet-4 and EfficientNet-B0 on FaceForensics++, evaluating both models across five practical dimensions.

Summary of Findings:

1. Both models are effective in-domain detectors: MesoNet-4 at AUC = 0.997, EfficientNet-B0 at AUC = 1.000.
2. EfficientNet-B0 is superior on hard, clean samples (ID AUC = 0.999) but suffers severe degradation under Gaussian noise (AUC = 0.555).
3. MesoNet-4 is substantially more computationally efficient (8.4× FLOPs, 143× parameters, 9.3× faster) and robust across all perturbation types.
4. No single model is comprehensively superior — the optimal choice depends on the deployment context.

Future Work

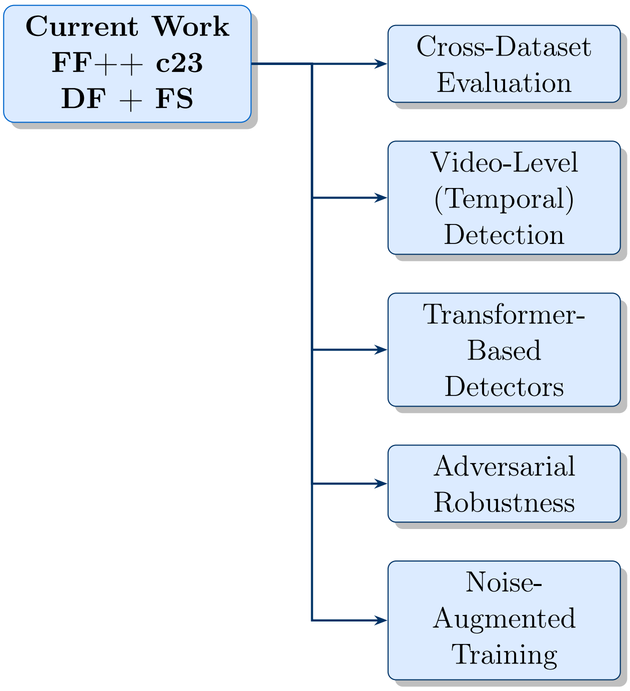


Figure 13: Proposed future directions extending this benchmark.

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